

Module 3

The Balance Sheet

Learning Objectives

By the end of this module you should understand:

1. The detailed structure of the balance sheet and the valuation problems surrounding fixed assets, inventories and debtors
2. The relationship between asset valuation and the measurement of profit
3. The impact of gearing on profits available for distribution to shareholders
4. Why a balance sheet must always balance
5. The significance, and construction, of the statement of cash flows

Agenda

3.1 Introduction

3.2 The Anatomy of the Balance Sheet

3.3 Fixed Assets

3.4 Current Assets

3.5 Current Liabilities

3.6 Net Current Assets and Net Assets

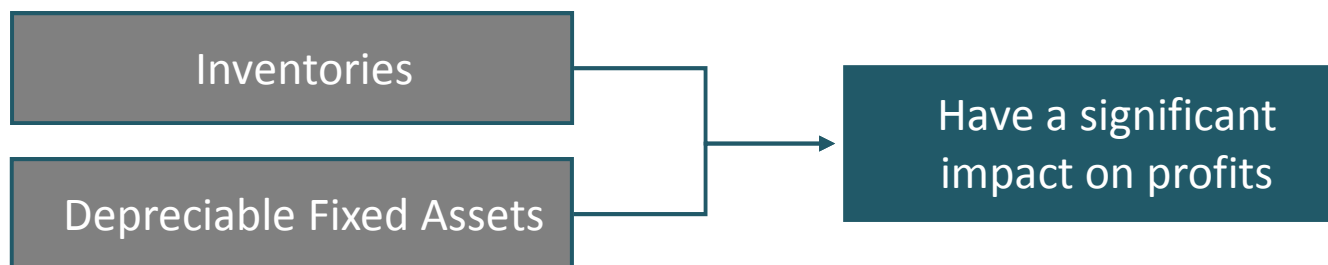
3.7 Why Does a Balance Sheet always Balance?

Review Questions

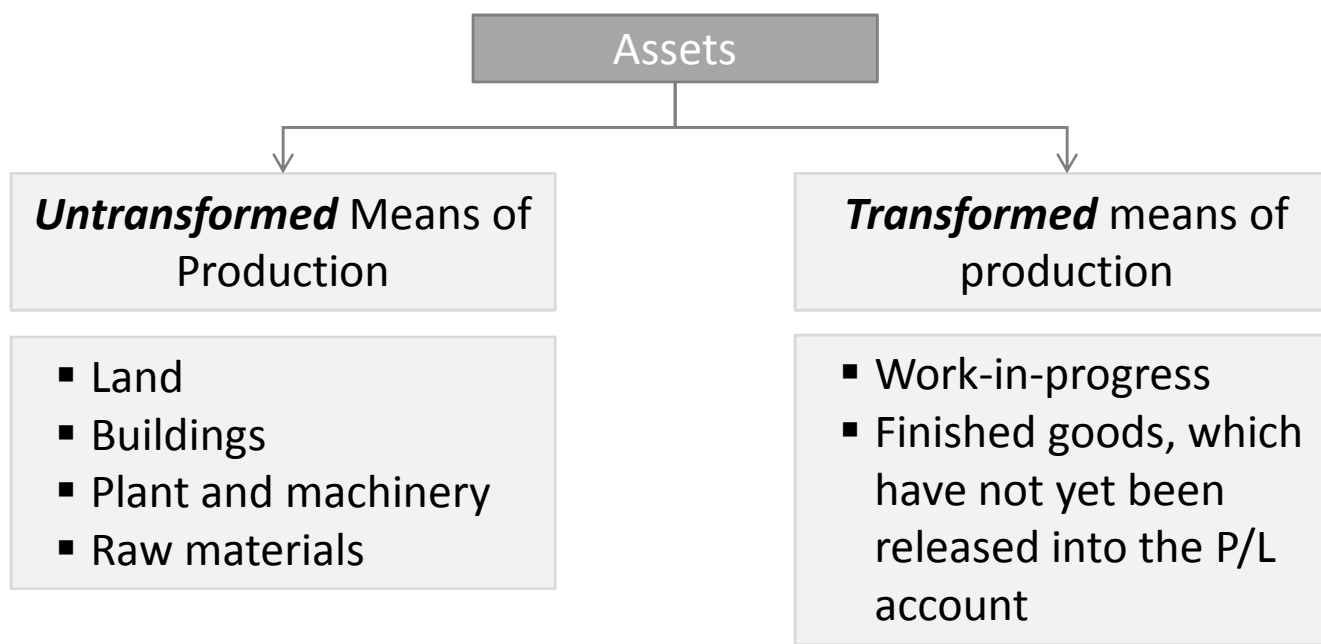
Case Studies

In module 2 we've seen that the measurement of a company's profit (or loss) depends to a very large extent on the valuation of certain assets

Valuation of assets like:



We've also seen that there are 2 categories of assets with varying values:



The following example shows how assets affect the profit and loss account

Problem

A new business purchases light fittings for its offices and warehouse for €1,000. During the first year it pays €100 to the local power utility company for energy consumed by these fittings. What are the accounting treatments for these transactions?

Solution

€100 energy consumption	Expense in the P/L Account	→ It has no future benefit
€1,000 Light Fittings	Fixed Asset in the Balance Sheet	→ It provides future benefit to the business
Annual Depreciation Cost	Expense in the P/L Account	→ Recognition of a loss in value
Annual Energy Costs	Expense in the P/L Account	→ Cost of power of the light fittings

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We will use the balance sheet of the sole trader encountered in Module 1 to discuss the balance sheet comprehensively

Balance sheet as at Action 7

Fixed assets		
Plant and equipment at cost	€ 12,000	
<i>Less: Depreciation</i>	50	€ 11,950
Current assets		
Inventories	5,500	
Debtors	750	
Cash	4,960	
	11,210	
<i>Less: Current liabilities</i>		
Creditors	3,000	8,210
Net assets of the company		20,160
Represented by:		
Capital introduced	20,000	
Profits earned	€ 160	
Owners' equity		20,160

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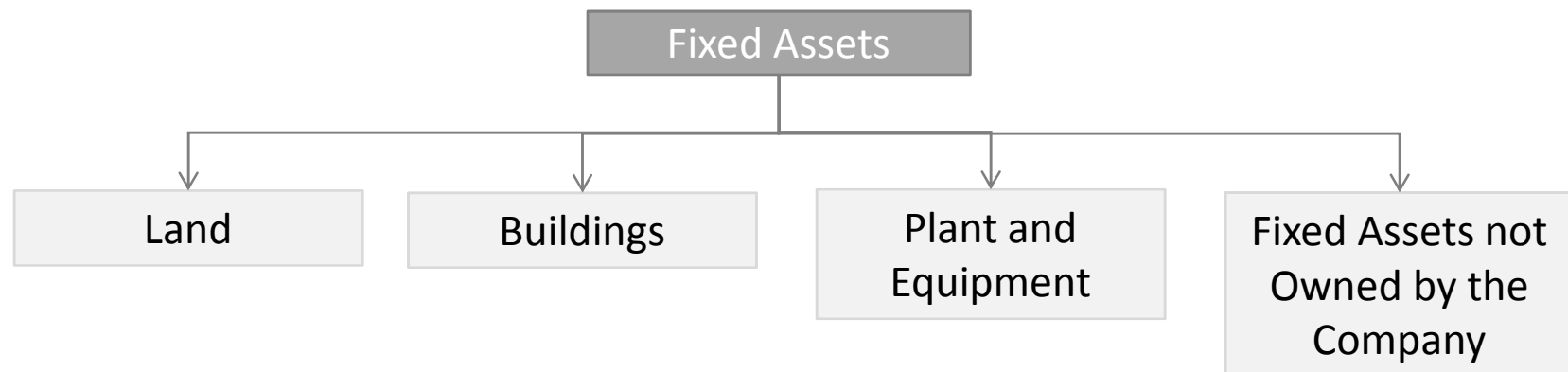
Review Questions

Case Studies

“Fixed assets” is the first item to be examined in the balance sheet

Fixed Assets definition

Assets that a company keeps for a **long** period of time, **not for resale** and to be used in the course of business



To classify an asset as a fixed asset, we have to consider how the management intends to use it

Classifying assets into fixed assets depends on:

The *intention* of management as to how to use the asset

NOT

Its physical characteristics

Example: Motor Vehicle can be treated in two different ways

Fixed Asset



Manufacturing
Company

Current Asset



Car Dealer

Land is recorded at its acquisition cost together with legal fees and the cost of preparing the land for use

Land

A fixed asset that is not usually depreciated since it has an **infinite life** and thus does **NOT** normally diminish in value

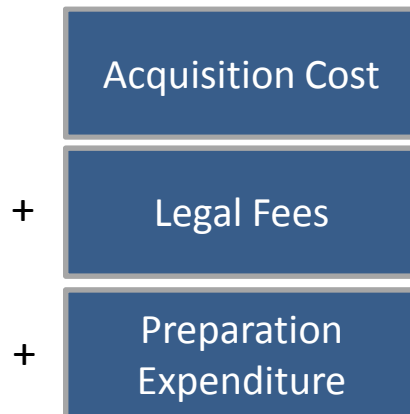
Expenditure related to the acquisition could be either:

Capitalized in the B/S

Or

Written Off in the P/L

Accountants record the asset at:



Any expenditure not related to the acquisition of the asset and making it operational

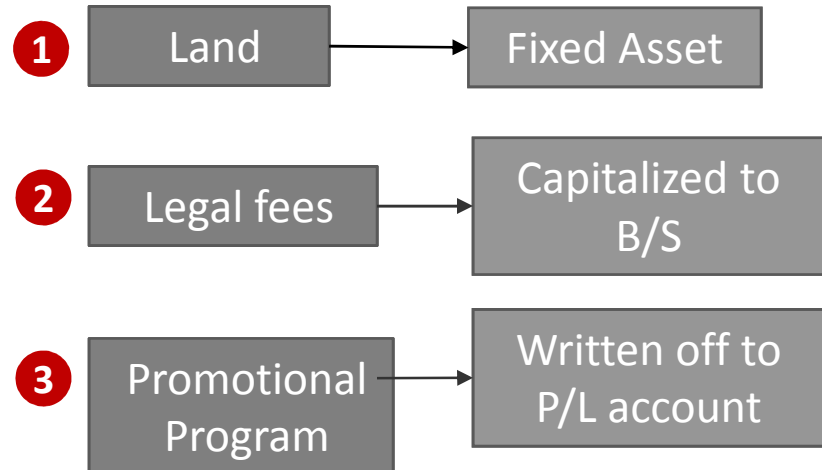
Accountants take a conservative position and write off expenditure unless it is clearly related to the acquisition of the asset

Problem

A retail grocery company purchased an urban site on which it proposes to locate a new supermarket within the next six months. The land cost €3.5 million, legal and survey fees cost €250,000 and a promotional program announcing the impending development involving billboards, handbills and local media advertising will cost another €250 000.

How much of this expenditure should be treated as a fixed asset?

Solution



Accountants are unhappy with the cost convention in relation to some assets, including land

That's because:

The acquisition cost is usually different from the current market value

2 Alternative Treatments:

1

To continue to record acquisition cost and to publish market value in parenthesis

2

To revalue the land from time to time, amend the book value (acquisition cost less depreciation to date) accordingly and add/subtract the incremental change to the shareholders' equity in the company

Should not be recorded in P/L

Market value is usually much greater than acquisition costs

Example

A company's land was revalued at the balance sheet date at €20,000. The land was purchased five years ago for €5,000 and has not been revalued until now.

Treatment 1: Balance sheet extract

<i>Fixed asset</i>	
Land	€5 000
(market value €20 000)	
Represented by:	
<i>Shareholders' equity</i>	
Owner's contribution	€X

Treatment 2: Balance sheet extract

<i>Fixed asset</i>	
Land	€20 000
Represented by:	
<i>Shareholders' equity</i>	
Owner's contribution	€X
Revaluation reserve	€15 000

*Profits of the
company are
unaffected by this
revaluation*

Not all accountants prefer using treatment 2

Some do prefer treatment 2

Argument:

They believe that land value gains will remain positive

They don't wait until the sale of the land to recognize these gains

Some don't prefer treatment 2

Argument:

The actual gain of €15,000 has not yet been realized

There is no intention to sell the land in the foreseeable future

The accounting treatment for buildings and plant & equipment differs from that of land

Buildings

*2 Accounting Treatments:
Either:*

Combined with Land

One Amount in the B/S

Or:

Depreciating asset

- Since they have a finite life
- The write-off cost is usually slow

Plant and Equipment

Accounting Treatment:

Depreciating asset

- As discussed in Module 2
- The slower depreciation rate, the smaller the impact of depreciation on profits

Many companies operate fixed assets they do not legally own – they do so under a leasing or rental arrangement

A company chooses to lease some assets, for a number of reasons:

- 1 Avoiding a substantial cash outflow at the beginning
- 2 Spreading relatively small cash outflows over asset's life
- 3 Replacing the leased asset with an up-to-date model without recording a loss/gain on sale of the old model
- 4 Saving further cash outflow as asset's maintenance costs are covered by the lease agreement
- 5 Lease payments are an allowable charge against profits before the calculation of taxation

There are 2 types of leases, each has its own accounting treatment

Finance Lease

Operating Lease

Definition

A lease in which ownership will revert to the lessee at the end of the lease

A lease in which ownership remains with the lessor

Balance Sheet Treatment

Capitalized

Not capitalized

Accounting Treatment

Both the capital and interest elements of the operating lease payments are charged in the P/L account

Only the interest element is charged while the capital portion is deducted from the lease obligation amount in the B/S

Long-term lease (finance lease) contract is in effect a purchase transaction and should be recorded in the balance sheet

The effect of recording the leased asset in the balance sheet can be seen from the following example

Example

Two shipping firms, Dee Ltd and Don Ltd have identical balance sheets.

Each firm decides to acquire a new tanker costing €10m. Dee Ltd will purchase its tanker with a loan from the bank and Don Ltd will lease its tanker.

Balance Sheet of Both Before Lease Agreement:

	€m		€m
Owner's equity	5	Total assets	10
Loans	5		
	10		10

Assets leased (finance lease) are shown in the balance sheet under 'fixed assets' and the future lease payments under 'creditors'

The respective balance sheets will look like this after the acquisition:

Dee Ltd

	€m		€m
Owner's equity	5	Total assets	20
Loans	15		
	<u>20</u>		<u>20</u>

Don Ltd

	€m		€m
Owner's equity	5	Assets	10
Loans	5	Leased assets	10
Creditors: lease obligations	10		
	<u>20</u>		<u>20</u>

This treatment relates company's earnings in an accounting period with the underlying assets which gave rise to the earnings

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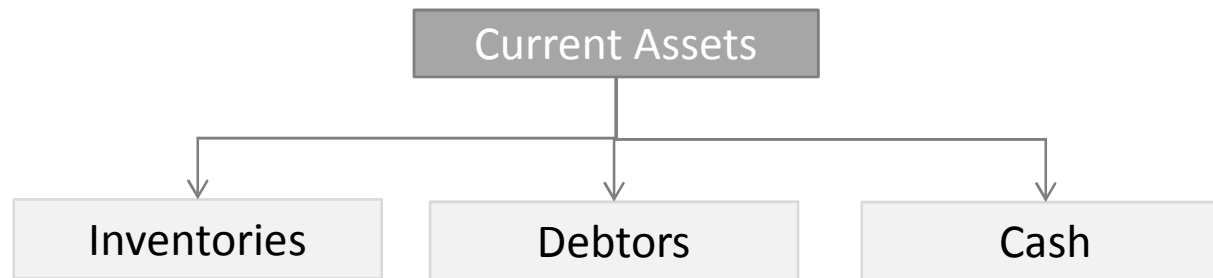
Review Questions

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The second category we examine in balance sheets is “current assets”

Current Assets definition

Assets which are expected to be sold or consumed during the normal operating cycle of the company, usually one year



The valuation of the three categories of inventories was discussed at length in Module 2

Inventory

1

Valuation of inventories is based on the 'lower of cost or market value' rule

2

'Cost' is usually defined as direct manufacturing cost plus a share of manufacturing overhead

'Period costs' are not normally allocated to inventory

3

Level of reported profit can be influenced by changing the basis of valuation for inventories

There is only one snag about the amount to record in the balance sheet for debtors: bad debt

Debtors

Customers who, at the end of the accounting period, have been billed by the company but who have not paid

Bad Debt Creates A Problem

Conservative Convention

Vs.

Matching Convention

Accountants like to write off the bad debt in the period which the debt is considered irrecoverable (which might be in another accounting period)

Expenses (bad debts) should be matched against the associated revenue

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graph TD; A[Conservative Convention] --> D[Provision for Bad Debts each period]; B[Matching Convention] --> D;
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Provision for Bad Debts each period

The P/L account is charged with an initial provision for bad debt

Bad Debt Treatment

1st Accounting Period

→ The P/L account is charged with an initial provision as if it represented real bad debts in the accounting period in which the provision is set up

2nd Accounting Period

→ The incremental adjustment in provision is entered in the P/L account

Management decides on the size of provision by considering a number of factors including:

- The risks attaching to each particular customer
- The severity with which the company will pursue recalcitrant debtors
- The general economic environment, e.g. interest rate changes

Through a provision for bad debts, a reduction in an asset "debtors" leads directly to a reduction in profits

Example

Year 1 Debtors (given) Policy: general provision for bad debts to be set at 5%	€100 000	Year 2 Debtors (given) Same policy on general provision	€122 000
		Specific bad debt arising in Year 2 to be written off	€2 000
Entry in profit and loss account Charge for bad debts provision	€5 000	Entry in profit and loss account Bad debt written off	€2 000
		Top-up provision to 5% of €120 000 Required €6 000 less existing €5 000	€1 000
Entry in balance sheet Debtors less provision	€95 000	Entry in balance sheet Debtors €120 000 Less: Provision €6 000	€114 000

The asset of cash is subject to least adjustment

Cash

Cash, whether it be located in the cash till, or departmental floats, or in the bank, can be counted

The managerial issue surrounding it:

Ensuring that the company has sufficient cash to meet its operating needs and investment plans

Idle cash produces no revenue

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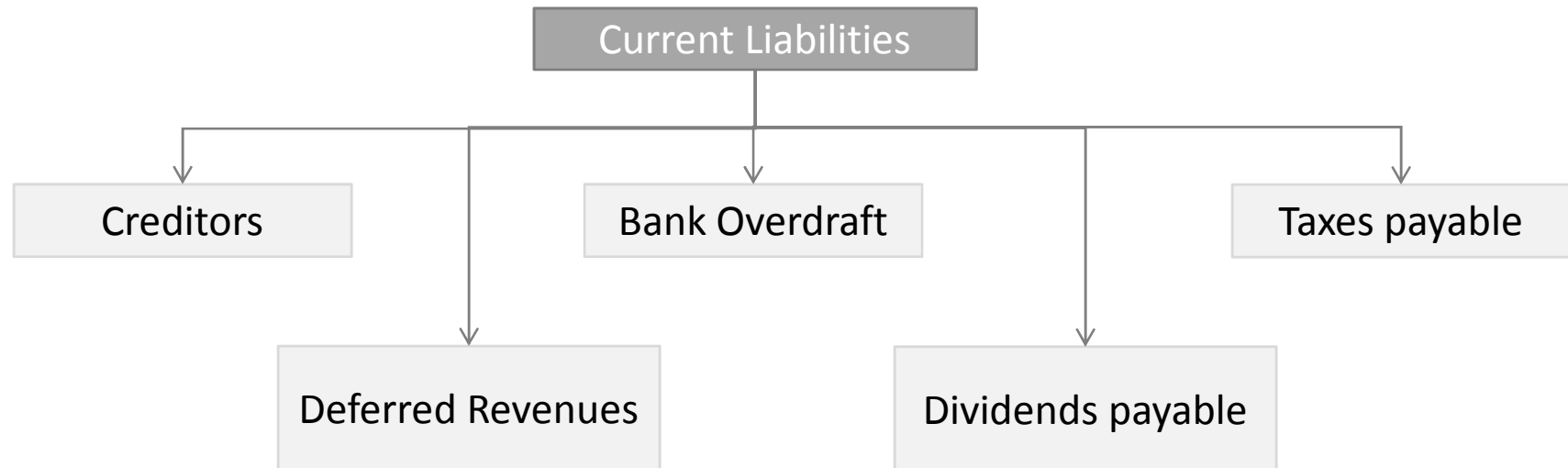
Review Questions

Case Studies

Current liabilities are those that a company expects to meet in the short term by paying them off in cash

Current Liabilities

Debts owed by the company which it expects to pay within the next 12 months



It is wrong to recognize deferred revenues as revenues in the P/L account of the period in which the payment has been received

Deferred Revenues

Revenue received by the company in advance of providing the goods or services

Example

A company which rents out factory space and receives from the tenant the rental in advance, this payment referring to part of the next accounting period



Example of Deferred Revenue

Example

In 20x0 the Aberdeen Investment Services Company launched a subscription newsletter which was intended to predict 'hot tips' on the UK Stock Exchange.

During its first two years of publication it enjoyed a considerable success rate in terms of the performance of shares selected and was voted 'Insider of the Year' in 20x1. The company therefore expected an increase in subscriptions for 20x2.

To take advantage of its reputation the company offered a three-year subscription starting in 20x2 for €210. The annual subscription remained at €75.

The entire amount of €210 had to be paid before the customer was placed on the mailing list. In 20x2 the company received 3000 one-year subscriptions and 2000 three-year subscriptions.

The deferred revenue is recognized in the P/L account in the year that the service is provided

Isolating this event, the financial statements for 20x2, 20x3 and 20x4 would appear as follows:

Balance sheets	20x2	20x3	20x4
Cash	€645 000	–	–
Owner's equity	€X	€X	€X
Deferred revenue	280 000	140 000	–
Profit and loss account			
Subscription revenue:			
One year	€225 000	–	–
Three years	140 000	€140 000	€140 000
	€365 000	€140 000	€140 000

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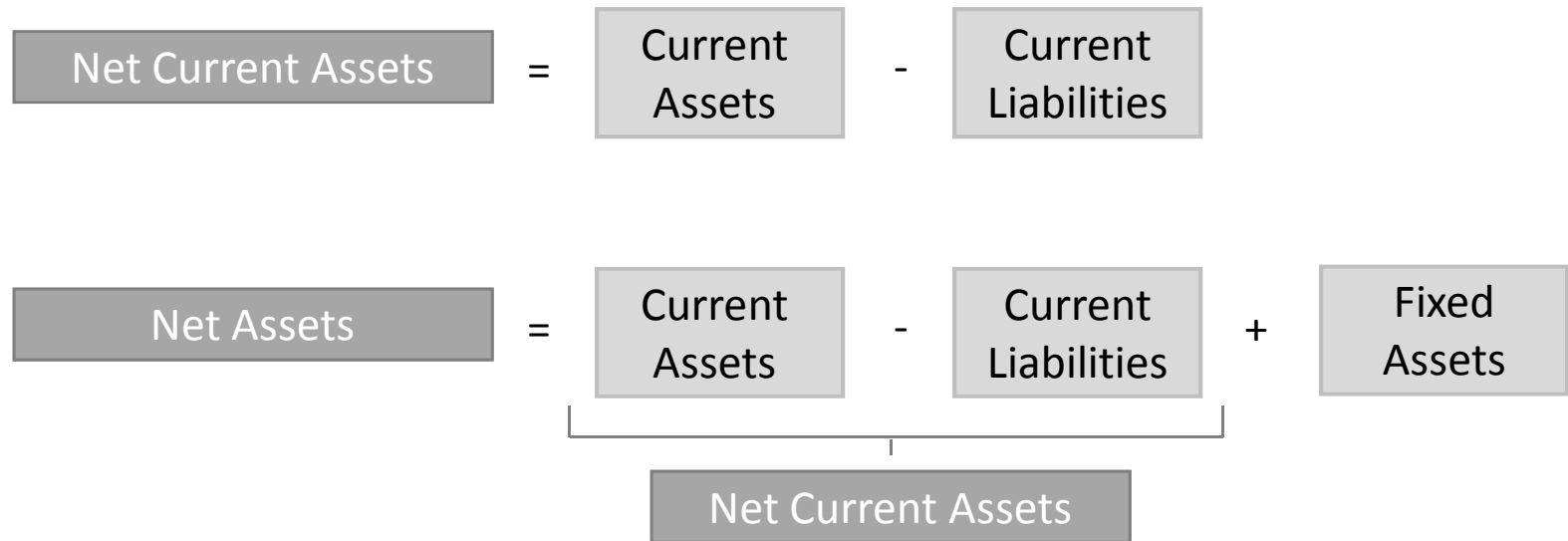
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Review Questions

Case Studies

Net current assets indicate a basic ability to meet liquid commitments from liquid resources



We will use the balance sheet of the sole trader encountered in Module 1 to further illustrate net assets and net current assets

Balance sheet as at Action 7

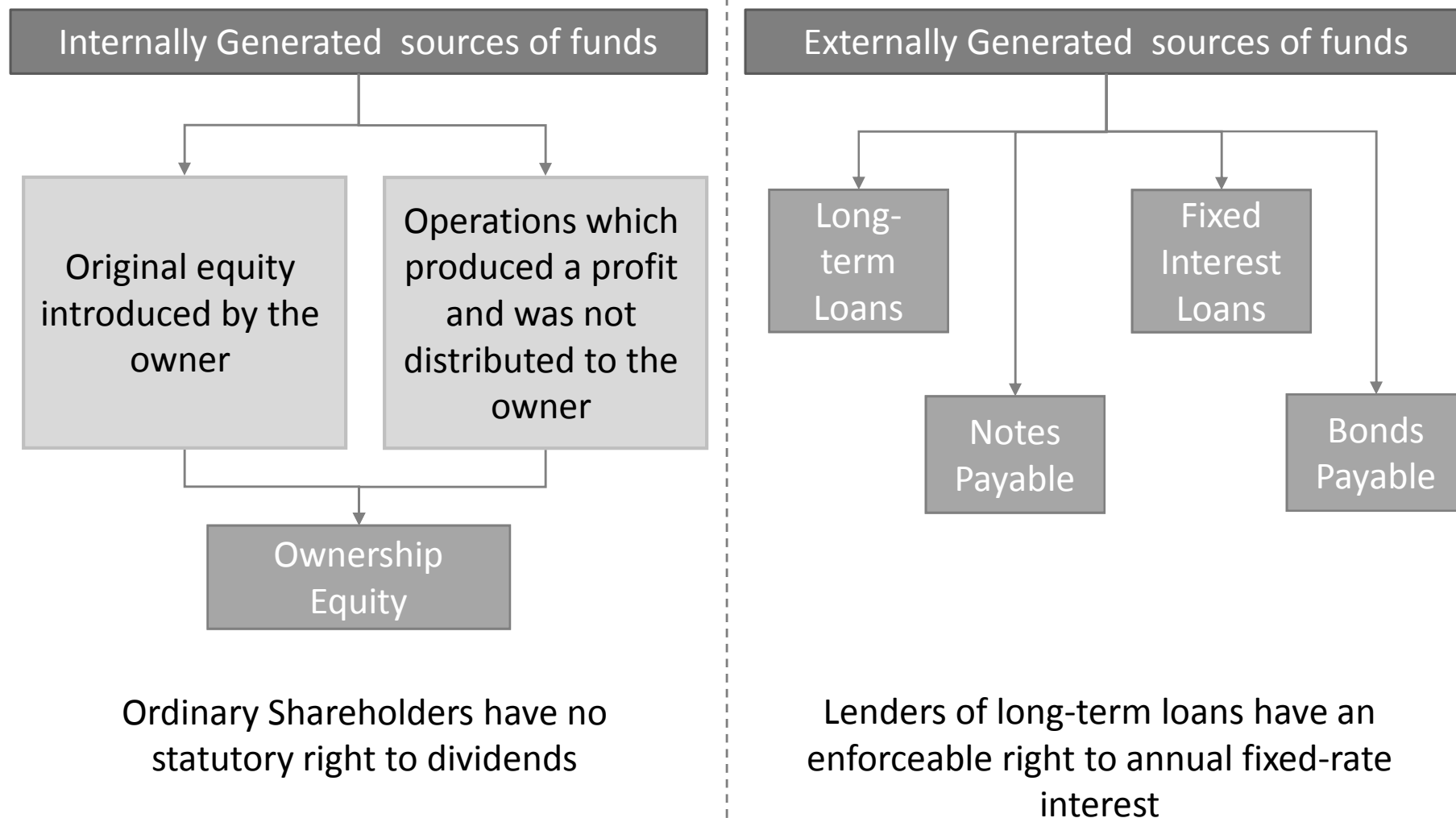
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<i>Less: Depreciation</i>	50	€ 11,950
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	11,210	
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Net assets of the company		20,160
Represented by:		
Capital introduced	20,000	
Profits earned	€ 160	
Owners' equity		20,160

Net Current Assets

Net Assets

Net assets can be financed either internally or externally

Financing Net Assets



Successful businesses would expect to see Owner's Equity build up year by year through profits earned but not distributed to owners

Example

Imagine that Actions 1–7 in the example in Module 1 represent one year. Imagine also that the profit and distributions to the owner are known for each of the next three years.

The build-up in owner's equity can be calculated as follows:

Year	Profits	Distribution to owners	Transfer to reserves
1	€160	–	€160
2	250	50	200
3	370	70	300
4	490	150	340
			1 000
Original capital introduction			20 000
Owners' equity at end of Year 4			€21 000

The relationship between shareholder funding (owners' equity) and loans is called gearing or leverage

Example

Two companies each have assets totaling €10,000 funded as follows:

	Company A	Company B
Owners' equity	€4 000	€8 000
10% long-term debt	6 000	2 000
	€10 000	€10 000
Gearing $\left(\frac{\text{Long-term debt}}{\text{Total assets}} \right)$	60%	20%
	↓	↓
	High-Geared Company	Low-Geared Company

In years of healthy profits, shareholders will achieve a better return on their money in a high-g geared company and vice versa

The effect of gearing on the return on owners' equity can be seen:

	Year 1 €	Year 2 €	Year 3 €
Profit before interest	1 000	600	1 500
Company A			
Interest on debt	600	600	600
Available to equity	400	–	900
	1 000	600	1 500
Return on equity:			
Company A (highly geared)	10%	–	22.5%
Company B			
Interest on debt	200	200	200
Available to equity	800	400	1 300
	1 000	600	1 500
Return on equity:			
Company B (lowly geared)	10%	5%	16.25%

$$\frac{\text{Available to Equity } \text{€}900}{\text{Equity } \text{€}4,000}$$

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There are many items on the asset side that are susceptible to multiple valuation while there are other items that are absolutely fixed

Variable Items

Fixed assets can be depreciated at many rates and by many methods

Inventories can be valued by many methods and may include varying proportions of factory overheads

Debtors may be reduced by whatever amount management considers to be prudent for a provision against bad debts

Fixed Items

Cash

Creditors

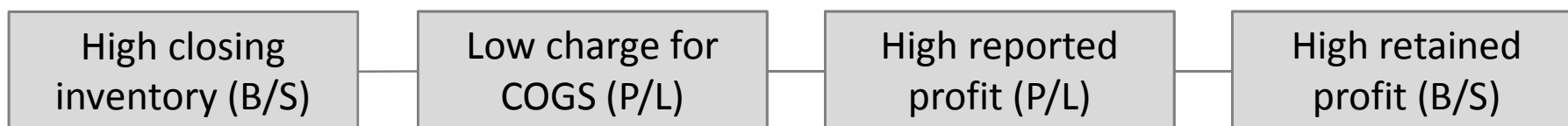
Long-Term Loans

Given this array of possible valuations it should be hard to always balance the balance sheet

The secret lies in understanding the role of the P/L account and how it relates to the balance sheet

Example

- Consider the situation of a management facing the decision to switch their inventory valuation from LIFO to FIFO
- The economic environment is inflationary and unit prices of the inventory items are rising
- The Knock-on effect of such a switch can be illustrated below:



Let's look at an example from module 2 that shows the relation between assets in the balance sheet with profit in the P/L account

Example

Altogether 1,200 units were purchased in a period of rising raw material prices at two monthly intervals as follows:

February	200 Units	@ €1 each	€200
April	200 Units	@ €1.5 each	300
June	200 Units	@ €2 each	400
August	200 Units	@ €2.5 each	500
October	200 Units	@ €3 each	600
December	200 Units	@ €3.5 each	700
	1,200 Units		2,700

The opening inventory of 100 units was valued at €1 per unit. Therefore the total costs to be allocated amount to €2800.

The P/L account acts as a kind of trampoline into which are dropped changes in asset valuations and out of which will bounce a matching impact on profits

	Under LIFO	Under FIFO
Sales:		
1000 units @ say €4.00	€4 000	€4 000
Opening stock	€ 100	€ 100
Purchases:		
1200 units @ varying prices	2 700	2 700
	<u>€2 800</u>	<u>€2 800</u>
Less: Closing stock 300 units	350	1 000
Cost of goods sold	<u>2 450</u>	<u>1 800</u>
Gross profit	<u>€1 550</u>	<u>€2 200</u>
Valuation of closing stock	LIFO	FIFO
300 units		
100 units @ €1.50 =	€150	100 units @ €3.00 = €300
200 units @ €1.00 =	€200	200 units @ €3.50 = €700
	<u>€350</u>	<u>€1 000</u>

Assuming no tax or distribution to shareholders:

Balance sheet under LIFO

Owner's equity increased by	€1550	Inventory valuation	€350
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Balance sheet under FIFO

Owner's equity increased by	€2200	Inventory valuation	€1000
(an increase over LIFO of €650)		(an increase over LIFO of €650)	

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Multiple Choice Questions

3.1 The following items appear in balance sheets.

- i. Cash, as current assets.
- ii. Inventories, as fixed assets.
- iii. Plant, as current assets.
- iv. Creditors, as current liabilities.

Which of the following is correct?

- A. (i) and (ii) only.
- B. (i) and (iv) only.
- C. (i), (ii) and (iii) only.
- D. (i), (ii) and (iv) only.

Multiple Choice Questions (Continued)

3.2 Companies acquire fixed assets for a variety of reasons, including:

- i. to act as a hedge against inflation;
- ii. to resell at a profit in the future;
- iii. to avoid cash surpluses;
- iv. to use in the course of the business.

Which of the following correctly reflects their primary reason(s)?

- A. (i) only.
- B. (i) and (ii) only.
- C. (iii) and (iv) only.
- D. (iv) only.

Multiple Choice Questions (Continued)

3.3 To accountants, capitalizing expenditure and writing off expenditure have identical impacts on the profit and loss account. True or false?

False

Multiple Choice Questions (Continued)

The following information applies to Questions 3.4–3.6.

A major electronics company has acquired a greenfield site to build a new factory. The land cost €1.5 million and there were professional fees of €100,000. The site requires remedial work costing €500,000 on old mineshafts before any building can commence. The company plans a massive advertising drive to promote the factory and to recruit skilled personnel. An initial package of €400,000 has been agreed with their advertising agents for this purpose.

3.4 How much of this expenditure will be written off in the profit and loss account?

- A. €100,000.
- B. €400,000.
- C. €500,000.
- D. €900,000.

Multiple Choice Questions (Continued)

3.5 What is the cost of the land in the company's balance sheet?

- A. €1.5 million.
- B. €1.6 million.
- C. €2.0 million.
- D. €2.1 million.

	€m
Acquisition cost	1.5
Professional fees	0.1
Remedial work	0.5
	2.1

Multiple Choice Questions (Continued)

3.6 If, after a two-year property boom, the land was revalued at €2.9 million, how much might then be transferred to a revaluation reserve?

- A. €0.8 million.
- B. €0.9 million.
- C. €1.3 million.
- D. €1.4 million.

Multiple Choice Questions (Continued)

3.7 There are several benefits for companies in choosing to lease fixed assets, rather than purchase them outright, including:

- i. beneficial cash flow impact of smaller payments;
- ii. overstatement of profits;
- iii. improved operational efficiency of assets;
- iv. replacement of assets without generating gains or losses.

Which of the following is correct?

- A. (i) and (ii) only.
- B. (i) and (iv) only.
- C. (ii) and (iii) only.
- D. (ii) and (iv) only.

Multiple Choice Questions (Continued)

3.8 The accounting profession has developed guidelines to deal with leased assets. In which of the following contractual agreements do these guidelines apply?

- A. Contract hire, for a vehicle.
- B. Rental, for a van.
- C. Finance lease, for a photocopier.
- D. Short-term hire, for a compressor.

Multiple Choice Questions (Continued)

3.9 The exclusion of leased assets from a balance sheet has the effect of:

- A. overstating both assets and liabilities;
- B. understating assets, but overstating liabilities;
- C. understating both assets and liabilities;
- D. overstating assets, understating liabilities.

Multiple Choice Questions (Continued)

3.10 Which of the following is NOT a current asset?

- A. Cash.
- B. Inventories.
- C. Vehicles.
- D. Debtors.

Multiple Choice Questions (Continued)

The following information applies to Questions 3.11–3.14.

Thames Limited has listed the following amounts in its balance sheet:

Cash	€10 000	Debtors	€55 000
Buildings	€75 000	Plant	€30 000
Creditors	€25 000	Inventories	€60 000
Taxes payable	€15 000	Dividends payable	€20 000

3.11 What is the amount of total current assets?

- A. €125,000.
- B. €150,000.
- C. €180,000.
- D. €255,000.

$$\text{Cash €10 000} + \text{Debtors €55 000} + \text{Inventories €60 000} = \text{€125 000}$$

Multiple Choice Questions (Continued)

3.12 What is the amount of total fixed assets?

- A. €75,000.
- B. €105,000.
- C. €130,000.
- D. €190,000.

$$\text{Buildings €75 000} + \text{Plant €30 000} = \text{€105 000}$$

Multiple Choice Questions (Continued)

3.13 What is the amount of total current liabilities?

- A. €35,000.
- B. €40,000.
- C. €50,000.
- D. €60,000.

$\text{Creditors €25 000} + \text{Taxes payable €15 000} + \text{Dividends payable €20 000} = \text{€60 000}$

Multiple Choice Questions (Continued)

3.14 What is the amount of net current assets?

- A. €40,000.
- B. €45,000.
- C. €65,000.
- D. €95,000.

Total current assets (from 11 above)	€125 000
<i>Less:</i> Total current liabilities (from 13 above)	60 000
Net current assets	<u>€ 65 000</u>

Multiple Choice Questions (Continued)

The following information applies to Questions 3.15–3.17.

Humber Limited has extracted the following information for its financial year-ends 20x0–20x2:

	Debtors	Specific bad debts written off
20x0	€150 000	Nil
20x1	€180 000	€3 500
20x2	€210 000	€5 700

Its policy towards bad debts is to make a general provision of 4 per cent of debtors. Specific bad debts have already been written off during each year.

3.15 What is the amount of the provision for bad debts at the end of 20x1?

A. €7,200.

B. €10,700.

C. €13,200.

D. €16,799.

$$\begin{aligned}\text{General provision} &= 4\% \text{ of Debtors } €180\,000 \\ &= €7\,200\end{aligned}$$

Multiple Choice Questions (Continued)

3.16 What is the amount charged for bad debts in the profit and loss account in 20x2?

- A. €5,700.
- B. €6,900.
- C. €7,200.
- D. €8,400.

(i)	Specific bad debts written off		€5 700
(ii)	Increase in bad debts provision of 20x2		
	Provision 4% of debtors =	8 400	
	Less: 20x1 Provision (<i>see 15 above</i>)	<u>7 200</u>	
			<u>1 200</u>
			<u>€6 900</u>

Multiple Choice Questions (Continued)

3.17 What is the amount of debtors less provision for bad debts at the end of 20x2?

- A. €192,400.
- B. €194,400.
- C. €195,900.
- D. €201,600.

Debtors	€210 000
Less: Bad debt provision (4%)	8 400
	<u>€201 600</u>

Multiple Choice Questions (Continued)

The following information applies to Questions 3.18–3.19.

Gleneagles Golf Club Limited has a financial year which runs to 31 March each year. The annual subscription has been fixed at £500 for the years to 31 March 20x0, 20x1 and 20x2. At 31 March 20x0, ten members had paid their annual subscriptions in advance. At 31 March 20x1, all 400 members were fully paid up, but 20 had paid their subscriptions in advance for the year to 31 March 20x2.

3.18 What is the subscription revenue for the year to 31 March 20x1?

- A. €195,000.
- B. €200,000.
- C. €205,000.
- D. €210,000.

Subscription revenue for year to 31 March 20x1 = $400 \times €500 = €200\,000$

Multiple Choice Questions (Continued)

3.19 What is the amount of deferred revenue at 31 March 20x1?

- A. €Nil.
- B. €5,000.
- C. €10,000.
- D. €15,000.

Deferred revenue at 31 March 20x1 = $20 \times €500 = €10\,000$
(subscriptions paid in advance).

Multiple Choice Questions (Continued)

The following information applies to Questions 3.20–3.21.

The following balances have been extracted from the accounting records of Mersey Limited at 31 January 20x0.

Creditors	€20 000	Prepayments	€2 000
Taxes payable	€10 000	Accruals	€4 500
Loan stock	€40 000	Bank overdraft	€8 500
Debtors	€45 000	Leasing creditor	€9 000

Additional information is available:

1. The leasing creditor comprises a further three years of lease payments of £3000 per annum.
2. The loan stock is repayable in four equal annual installments, the first of which is due on 31 January 20x1.

Multiple Choice Questions (Continued)

3.20 What is the amount of total current liabilities at 31 January 20x0?

- A. €43,000.
- B. €46,000.
- C. €52,000.
- D. €56,000.

Current liabilities	€
Creditors	20 000
Taxes payable	10 000
Accruals	4 500
Bank overdraft	8 500
	43 000
<i>Add:</i>	
Current portion of long-term liabilities	
Loan stock ($\frac{1}{4} \times €40\,000$)	10 000
Leasing creditor ($\frac{1}{3} \times €9\,000$)	3 000
	56 000

Multiple Choice Questions (Continued)

3.21 What is the amount of total long-term liabilities at 31 January 20x0?

- A. €36,000.
- B. €39,000.
- C. €46,000.
- D. €49,000.

Long-term liabilities	€
Loan stock ($\frac{3}{4} \times \text{€}40\,000$)	30 000
Leasing creditor ($\frac{1}{3} \times \text{€}9000$)	6 000
	36 000

Multiple Choice Questions (Continued)

3.22 In preparing year-end financial statements, which of the following is the correct treatment for deferred revenue?

- A. As a current asset in the balance sheet.
- B. As a current liability in the balance sheet.
- C. As a long-term liability in the balance sheet.
- D. As an expense in the profit and loss account.

Multiple Choice Questions (Continued)

3.23 In more profitable years, ordinary shareholders can expect a higher return on their investment than lenders of long-term loans. True or false?

True

Multiple Choice Questions (Continued)

3.24 Which of the following costs will normally be excluded from the valuation of inventories?

- A. Direct materials.
- B. Variable manufacturing overhead.
- C. Selling and distribution overhead.
- D. Fixed manufacturing overhead.

Multiple Choice Questions (Continued)

The following information applies to Questions 3.25–3.26.

Solway Limited has the following three-year record of profits before interest:

Year	€
1	55 000
2	36 000
3	48 000

Assume that owner's equity is fixed at €100,000 for each of the three years. The company has been offered long-term loan stock at 10 per cent per annum.

3.25 In Year 2, how much long-term loan stock would result in a return on owner's equity of 25 per cent?

- A. €110,000.
- B. €230,000.
- C. €300,000.
- D. €350,000.

Multiple Choice Questions (Continued)

3.26 In Year 3, how much long-term loan stock would result in a gearing ratio of 37.5 per cent?

A. €37,500.

B. €60,000.

C. €137,500.

D. €160,000.

$$\text{Gearing} = \frac{\text{Long-term debt}}{\text{Total assets}} = 37.5\%$$

$$\begin{aligned}\text{Substitute} &= \frac{x}{100\,000 + x} = 37.5\% \\ &= €60\,000\end{aligned}$$

Agenda

3.1 Introduction

3.2 The Anatomy of the Balance Sheet

3.3 Fixed Assets

3.4 Current Assets

3.5 Current Liabilities

3.6 Net Current Assets and Net Assets

3.7 Why Does a Balance Sheet always Balance?

Review Questions

Case Studies

Case Study 3.1

Clydemouth Chemicals Limited has purchased 1,500 kg of material Y over a six-month period, as follows:

	Quantity kg	Unit cost €	Total cost €
April	300	3.00	900
May	400	3.25	1 300
July	250	4.00	1 000
August	150	4.20	630
September	400	4.50	1 800
	<u>1 500</u>		<u>5 630</u>

At 1 April, the company had a stockholding of 120 kg, costing £336. Sales revenue for the six months to 30 September amounted to £8000, in respect of 1000 kg sold.

Required

Prepare profit and loss accounts for the six months, adopting:

- (a) LIFO inventory basis;
- (b) FIFO inventory basis

Outline the balance sheet implications of each basis.

Solution

	Under LIFO		Under FIFO	
	€	€	€	€
Sales		8 000		8 000
Cost of sales				
Opening stock	336		336	
Purchases	5 630		5 630	
	<u>5 966</u>		<u>5 966</u>	
Closing stock	<u>1 886 (1)</u>		<u>2 710 (2)</u>	
		4 080		3 256
Gross profit		<u>3 920</u>		<u>4 744</u>

Notes

(1) LIFO	– 620 kg		
	– applied to		
		$120 \times \text{€}2.80$	336
		$300 \times \text{€}3.00$	900
		$200 \times \text{€}3.25$	650
			<u>1 886</u>
(2) FIFO	– 620 kg		
	– applied to		
		$400 \times \text{€}4.50$	1 800
		$150 \times \text{€}4.20$	630
		$70 \times \text{€}4.00$	280
			<u>2 710</u>

Case Study 3.2

Longhorn Whisky Ltd and Inverness Distillers Ltd employ equal amounts of assets in their respective businesses. However, the funding of the assets differs, as follows:

	Longhorn Whisky Ltd	Inverness Distillers Ltd
	€	€
Owners' equity	40 000	80 000
Long-term debt	60 000	20 000
	100 000	100 000

The long-term debt in each company attracts an interest charge of 8 per cent per annum. For three years, both companies have earned the same operating profits (before interest):

	€
Year 1	10 000
Year 2	6 000
Year 3	15 000

Required

Compute the gearing ratios for each company and assess the impact of the gearing on the return on equity for each company in each year.

Solution

	Longhorn Whisky Ltd	Inverness Distillers Ltd
Gearing % (Long-term debt/ Total assets)	€60 000/€100 000 = 60%	€20 000/€100 000 = 20%

Longhorn Whisky Ltd			
	Year 1	Year 2	Year 3
	€	€	€
Operating profits (before interest)	10 000	6 000	15 000
<i>Less:</i>			
Interest on long-term debt 8% × €60 000	4 800	4 800	4 800
Profits available to equity	<u>5 200</u>	<u>1 200</u>	<u>10 200</u>
Return on equity %			
Profit available to equity / Equity	5 200/40 000	1 200/40 000	10 200/40 000
=	<u>13%</u>	<u>3%</u>	<u>25.5%</u>

Solution (Continued)

Inverness Distillers Ltd

	Year 1	Year 2	Year 3
	€	€	€
Operating profits (before interest)	10 000	6 000	15 000
<i>Less:</i>			
Interest on long-term debt $8\% \times \text{€}20\,000$	1 600	1 600	1 600
Profits available to equity	<u>8 400</u>	<u>4 400</u>	<u>13 400</u>
Return on equity %	$8\,400/80\,000$	$4\,400/80\,000$	$13\,400/80\,000$
	<u>= 10.5%</u>	<u>5.5%</u>	<u>16.75%</u>

Longhorn is more highly geared than Inverness with 60 per cent gearing as against 20 per cent. In times of higher profits (Years 1 and 3), Longhorn's gearing produces a higher return on equity. However, when profits are lower (Year 2), the high gearing works to the disadvantage of the equity interests in Longhorn.